# MRI COIL LOADING (PATIENT EFFECT) – UNDERSTANDING THE IMPACT

GE MRI

The patient becomes part of the RF system and changes the coil's impedance and tuning.

Good Loading → Tune & Match Easily → High SNR & Uniform Images | Poor Loading → Detune & Mismatch → Signal Loss & Artifacts

## 1. WHAT IS COIL LOADING?

When a patient is placed in the coil, their body (high water content) interacts with the RF magnetic field. This causes:

- Resonant frequency shift ($f_0$)
- Change in impedance (Z)
- Change in Q (quality factor)
- Change in SNR & penetration

The patient is part of the circuit!
Always tune & match with the patient in the coil whenever possible.

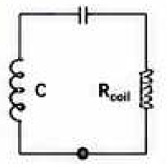
## 2. HOW LOADING CHANGES THE COIL

| Parameter | Without Patient (Unloaded) | With Patient (Loaded) | Effect |
|---|---|---|---|
| Resonant Frequency ($f_0$) | Higher | Lower | ↓ |
| Impedance (Z) | Higher | Lower | ↓ |
| Quality Factor (Q) | Higher | Lower | ↓ |
| Bandwidth (BW = $f_0$ / Q) | Narrower | Wider | ↑ |
| SNR | Lower | Higher | ↑ |
| Penetration | Lower | Higher | ↑ |

Typical frequency shift with body coil:
–1 to –3 MHz (on 64–128 MHz systems)

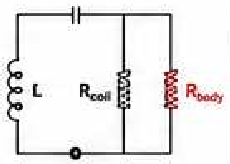
## 3. EQUIVALENT CIRCUIT VIEW

**UNLOADED COIL (No Patient)**




High $f_0$, High Z, High Q
Narrow BW

**LOADED COIL (With Patient)**




Lower $f_0$, Lower Z, Lower Q
Wider BW (more damping)

$R_{body}$ represents the patient (lossy tissue) that absorbs energy and lowers Q and impedance.

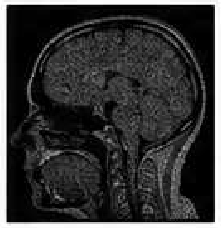
## 4. LOADING EFFECT ON SNR & PENETRATION

**GOOD LOADING (Well matched & tuned)**

- Efficient power transfer
- High SNR
- Better penetration
- Uniform images

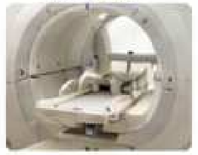
**POOR LOADING (Detuned / Mismatched)**

- Poor power transfer
- Low SNR
- Shallow penetration
- Non-uniform / noisy images



## 5. LOADING EXAMPLES – GE COILS

GE HD Body Coil (Integrated)



GE 8ch Head Coil (Receive Only)

GE Spine Array (16 Channel)

GE Knee Coil (8 Channel)

Different coils load differently depending on size, elements, and patient size/position.

## 6. TYPICAL LOADING EFFECTS (VALUES VARY BY SYSTEM & COIL)

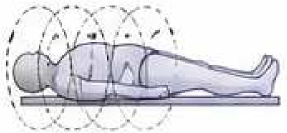
| Coil Type | Frequency Shift (Δf) | Impedance Change (ΔZ) | Q Change | Comments |
|---|---|---|---|---|
| Body Coil (Integrated) | –1 to –3 MHz | ↓ 30–60% | ↓ 2–5x | Large loading, more damping |
| Head Coil (8–32 ch) | –0.5 to –2 MHz | ↓ 20–50% | ↓ 2–4x | Moderate loading |
| Spine Coil (Array) | –0.5 to –1.5 MHz | ↓ 20–40% | ↓ 1.5–3x | Depends on FOV & patient size |
| Extremity Coil (Knee/Shoulder) | –0.2 to –1 MHz | ↓ 10–30% | ↓ 1.2–2x | Less loading |

↑ = Increase ↓ = Decrease

## 7. LOADING VS PATIENT SIZE / POSITION

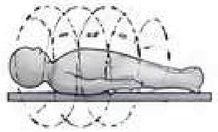
**LARGER PATIENT / MORE IN COIL**

- More tissue in the RF field
- More loading (lower $f_0$, lower Z, lower Q)
- Higher SNR but more absorption



**SMALLER PATIENT / OFF-CENTER**

- Less tissue in the RF field
- Less loading ($f_0$ closer to unloaded)
- Lower SNR, possible hot spots



Always center the patient and use proper coil selection.
Correct positioning = consistent loading = consistent image quality.

## 8. HOW TO MANAGE LOADING

- Tune with patient in the coil.
- Match with patient in the coil.
- Use correct coil for exam & patient size.
- Center patient in coil (left-right, up-down).
- Use proper FOV & prescription.
- Re-tune if patient position changes.
- Use pre-scan normalization (Auto PNS, etc.).

## 9. SIGNS OF POOR LOADING

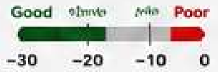
- Large frequency shift (> 2–3 MHz)
- Low impedance (< 10 Ω at 50 Ω system)
- Very low Q (< 10)
- Reflected power high (> –10 dB)
- Low SNR / non-uniform images
- Bands / shading / folds in image

Example: Poor Loading

Check tune, match, position, coil, and FOV.

## 10. LOADING CHECK – WHAT ENGINEERS MONITOR

| Frequency (MHz) | Impedance (Ω) | | Q Factor (Δ) | Reflected Power (dB) |
|---|---|---|---|---|
| 62.70 | 52.3 | | 20.1 | –22.4 |
| Δf (MHz) –1.35 | R (Ω) 51.8 | X (Ω) 5.2 | | |
| Tuned | Matched | | Normal | Good |




Ideal Reflected Power: –20 dB or lower
(Lower is better)

## 11. COMMON LOADING PROBLEMS & CAUSES

| Problem | Possible Cause | What to Do |
|---|---|---|
| Large frequency shift | Wrong coil, off-center, large patient | Re-center, use correct coil, re-tune |
| Low impedance | Heavy loading, detuned, coil cable issue | Re-tune, check cables, inspect coil |
| Low Q | Over-loaded, lossy cable, coil not tuned | Re-tune, check coil & cables |
| High reflected power | Poor match, detuned | Re-match, check tune |
| Shading / bands | Off-center, poor loading, coil element failure | Re-position, coil check, run coil test |

## 12. BEST PRACTICES SUMMARY

- Tune with patient.
- Match with patient.
- Center patient.
- Use correct coil.
- Keep coil & connectors clean & dry.
- Check cables and element indicators.
- Document loading values if out of range.

## 13. QUICK CHECK FLOW

1. Tune (Δf ~ 0 MHz with patient)
2. Match (Z ~ 50 + j0 Ω)
3. Check Reflected Power (≤ –20 dB)
4. Check VSWR (≤ 1.5 : 1)
5. Check Q (10–30 typical)
6. Verify image quality

### ENGINEER NOTES

- Loading is patient, coil, position, and frequency dependent.
- Always tune & match after any patient change.
- Good loading = good images, short scans, happy patients.

### ABBREVIATIONS

| | | | |
|---|---|---|---|
| $f_0$ | = Resonant Frequency | Q | = Quality Factor |
| Z | = Impedance (Ω) | BW | = Bandwidth |
| R | = Resistance (Ω) | SNR | = Signal-to-Noise Ratio |
| X | = Reactance (Ω) | FOV | = Field of View |

### SAFETY REMINDER

- High RF power can cause tissue heating.
- Ensure patient comfort and monitoring.
- Follow all site safety procedures.

### THE BOTTOM LINE

Understand loading. Manage loading.
Deliver power efficiently.
Get the best SNR, uniformity, and diagnostic confidence.